Name the force and its source

antocciphysics | TEACHER KEY | CPA

Learning target: Identify forces on a chosen object and name each interaction partner.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 7; 3 course days later.*
Slowing down; change is 8 m/s east. Acceleration 2.667 m/s² east.

02. Explain a worked example (worked)
Weight by Earth down; support by table up. Paired force is gravity by the book on Earth, not the support force.

03. Which object gets the force? (model)
Earth exerts weight down; table exerts normal/support up. Equal magnitudes for rest.

04. Mass and weight (practice)
Weight 20 N down; rope tension 20 N up for rest.

05. After the throw (reason)
No hand contact; Earth's gravity/weight acts down. Upward velocity is not an upward force.

06. Choose a method (mixed) *Review: Unit 1, Day 9; 1 course days later.*
The straight-moving rover can have zero net force. The turning rover changes velocity direction and needs net force.

07. Exit check (exit)
Earth on lamp 30 N down; cord on lamp 30 N up.

08. STRETCH: Apply it another way (stretch)
Add downward hand force. Upward support increases to balance weight plus hand force.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 7; 3 course days later.*
First a=-5 m/s², speeding up. Then a=0, constant speed 24 m/s.

02. Explain a worked example (worked)
Weight by Earth down; support by table up. Paired force is gravity by the book on Earth, not the support force.

03. Inventory the interactions (model)
Earth: weight; floor: normal and friction; rope: tension. Directions depend on pull/contact, friction opposes relative slip.

04. Mass and weight in two places (practice)
29.4 N and 4.8 N downward; mass stays 3.0 kg.

05. Normal is not always weight (reason)
N+12-49=0; N=37 N. Rope supplies part of upward support.

06. Choose a method (mixed) *Review: Unit 1, Day 9; 1 course days later.*
Tangent velocity direction changes, so acceleration and net force are inward and nonzero. Constant speed is insufficient for equilibrium.

07. Exit check (exit)
Weight 19.6 N down by Earth; tension 19.6 N up by string. Object-on-string acts on string, not object.

08. STRETCH: Apply it another way (stretch)
Weight 20 N down, hand 15 N down, normal 35 N up. Partners: book pulls Earth up, pushes hand up 15 N and table down 35 N.